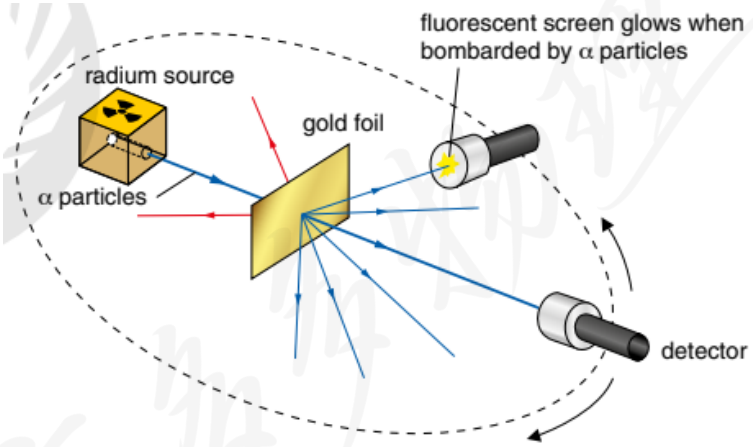
E2.2.1 Rutherford's model and its limitation

Rutherford directed alpha particles at very thin gold foil. Most passed almost straight through, showing that atoms are mostly empty space. Only a few were strongly deflected or reflected, showing that strong repulsion is concentrated in a tiny region. The model therefore places positive charge and almost all mass in a small nucleus, with electrons outside.

Thin foil reduces repeated scattering by several nuclei, so a deflection more clearly reflects one close encounter.

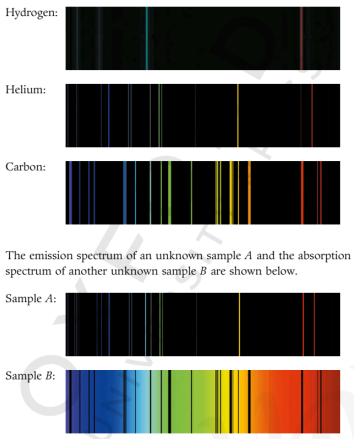
A classically orbiting electron accelerates and should continually radiate energy. This cannot explain atomic stability. The model also lacks discrete energy levels and therefore cannot explain spectra containing only specific wavelengths. Bohr's stationary states and transitions address these problems.



Most alpha particles pass through thin gold foil, while a few are deflected. The detector can be moved to measure scattering in different directions.

E2.2.2 Identifying elements from spectral lines

Compare an unknown sample's line wavelengths with known elemental patterns. Bright emission and dark absorption lines of the same element occur at corresponding wavelengths. Match the positions of the whole set rather than only line count or brightness. In the illustrated example, sample A's emission includes hydrogen and helium, while sample B's absorption matches carbon.



Match known elements and unknown samples at the same spectral-line wavelengths.

E2.2.3 Stationary states and angular momentum

Bohr's hydrogen model allows only stationary orbits with quantised angular momentum. Electrons do not radiate in stationary states. Transitions absorb or emit photons with energies equal to level differences. The quantum number is a positive integer, not a continuous variable. This hydrogen model cannot be extended unchanged to all many-electron atoms.

$$m_e v_n r_n = \frac{nh}{2\pi}$$

$$L = \frac{nh}{2\pi}, \quad n = 1, 2, 3, \ \ldots$$

Bohr hydrogen model: m_e: electron mass (kg); v_n: orbital speed (m s^{-1}); r_n: radius (m); n: positive integer; h: Planck constant (J s).

Bohr hydrogen model: electric attraction supplies centripetal force for circular orbits. Only discrete stationary orbits are allowed; an electron in one such orbit does not radiate and has constant energy. Transitions between orbits involve absorption or emission of a photon with the energy difference. Orbital angular momentum L (J s) is quantized; h = 6.63 × 10^{-34} J s.

E2.2.4 Orbital radii in hydrogen

In Bohr's hydrogen model, the radius of state n is proportional to n squared, so doubling the quantum number quadruples radius. Only positive integer n values are allowed and n = 1 is the ground state.

$$r_n = n^2 r_1$$

Bohr hydrogen atom: r_n: radius of orbit n; r_1: ground-state radius; n = 1, 2, 3, … .

Hydrogen energy levels

With zero energy assigned to a stationary electron at infinity, bound hydrogen levels are negative. The ground state n = 1 has the lowest energy and n = 2 is the first excited state. Larger n gives levels closer to zero and more closely spaced. Negative energy means binding. Required transition energy is final minus initial energy.

$$E_n = \frac{E_1}{n^2} = \frac{-13.6\ \text{eV}}{n^2}$$

Hydrogen atom; zero energy is a stationary free electron infinitely far from the nucleus. n: positive integer; E_1 = −13.6 eV.

Excitation energy

Excitation raises an electron from a lower to a higher bound level. The required energy is final minus initial energy. The displayed formula starts from the ground state; use the actual initial level for an already excited atom.

Photon absorption and electron impact have different energy conditions. A photon is absorbed as a whole in a bound-state transition and must match an allowed energy gap. An incident electron can transfer only the required part and leave with remaining kinetic energy.

Process	Energy condition	Result
Elastic collision	No energy enters internal excitation	Atom remains unexcited; a small recoil-energy transfer is possible
Excitation	Incident energy can supply an allowed level gap	A bound electron rises to a higher level; the incident electron retains excess energy
Ionisation	Incident energy is sufficient to remove a bound electron	The atom loses a bound electron; outgoing particles carry excess energy

$$E_{\text{exc}} = E_n - E_1$$

Excitation from ground state to bound level n > 1. Energy difference is positive; use consistent energy units.

Ionisation energy

Ionisation removes a bound electron from the atom. With electron energy zero at infinity, the minimum required energy is the negative of its initial bound energy. Ground-state hydrogen requires 13.6 eV; excited states require less.

An incident photon or electron can supply the ionisation energy, with excess energy becoming outgoing kinetic energy. After electron-impact ionisation, the atom is positively charged because it has lost a bound electron. The incident electron can also leave.

$$E_{\text{ion}} = E_{\infty} - E_1 = 13.6\,\text{eV}$$

Hydrogen atom initially in its ground state; E∞ = 0. From excited level n, ionization requires 13.6/n² eV.

State	Quantum number	Energy level	Radius of orbit
ground state	$n = 1$	$E_1 = -13.6$ eV	$R_1 = 5.29 \times 10^{-11}$ m
1st excited state	$n = 2$	$E_2 = -\frac{13.6}{2^2} = -3.40$ eV	$R_2 = (2)^2(5.29 \times 10^{-11}) = 2.12 \times 10^{-10}$ m
2nd excited state	$n = 3$	$E_3 = -\frac{13.6}{3^2} = -1.51$ eV	$R_3 = (3)^2(5.29 \times 10^{-11}) = 4.76 \times 10^{-10}$ m
3rd excited state	$n = 4$	$E_4 = -\frac{13.6}{4^2} = -0.85$ eV	$R_4 = (4)^2(5.29 \times 10^{-11}) = 8.46 \times 10^{-10}$ m

Table 2.2b *Energy levels and radii of orbits in different states.*

As quantum number n increases, bound-state energy approaches zero from below and orbit radius increases. The energy spacing becomes smaller.

E2.2.5 Energy transitions and spectral wavelengths

An electron dropping to a lower level emits a photon equal in energy to the level difference. A larger difference means higher frequency and shorter wavelength. Use hydrogen energy levels in the correct order to obtain positive photon energy. With SI Planck constant, convert level differences from eV to J first.

$$hf = \frac{hc}{\lambda} = E_a - E_b$$

$$\frac{1}{\lambda} = \frac{13.6\,\text{eV}}{hc}\left(\frac{1}{b^2} - \frac{1}{a^2}\right)$$

Emission from higher level a to lower level b, a > b. Photon energy is positive. Convert eV to J when using h in J s; λ is vacuum wavelength.

Hydrogen emission a > b. Convert 13.6 eV to joules if h is in J s and c in m s⁻¹; then λ is in metres.

E2.2.6 Photon momentum

A photon has no rest mass but carries energy and momentum. Its momentum magnitude is inversely proportional to vacuum wavelength and directed along propagation. Shorter-wavelength photons have greater momentum and can transfer it through absorption or reflection. Substituting zero rest mass into the ordinary non-relativistic mass-times-velocity formula does not give photon momentum.

$$p = \frac{h}{\lambda}$$

Photon momentum p (kg m s^{-1}); h = 6.63 × 10^{-34} J s; λ: vacuum wavelength (m).

E2.2.7 The de Broglie wavelength

Matter particles have wavelengths associated with momentum: larger momentum gives shorter wavelength. Replace momentum by mass times speed only in the non-relativistic regime. Electron wavelengths can show observable wave effects at suitable scales, while macroscopic wavelengths are tiny. Accelerating electrons raises momentum, shortens wavelength and improves the wavelength-related diffraction limit.

$$\lambda = \frac{h}{p} \approx \frac{h}{mv}$$

de Broglie wavelength λ (m); momentum p (kg m s^{-1}); h = 6.63 × 10^{-34} J s. p ≈ mv only at non-relativistic speed.